SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

CO2 ASSESSMENT TOOL 2 – Architecture Design Assignment

Course Code:	CSA1007	Course Title:	Software Engineering
CO Assessed:	CO2	Assessment:	Assessment Tool 2 – Architecture Design Assignment
Weightage:	20%	Total Marks:	25
CO2 Statement:	*Perform detailed requirements analysis, design scalable architectures, and prioritize features with a focus on cross-disciplinary skills*		
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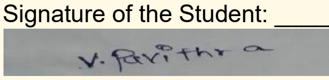
Date:	30/07/2026	Duration:	60 minutes

Code of Conduct

I____V.PAVITHRA_ 192524274__________ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _________________________



Annexure A – Architecture Design Assignment

CO 2 ASSESSMENT TOOL 2
ARCHITECTURE DESIGN ASSESSMENT

AUTONOMOUS ELECTRIC VEHICLE CHARGING NETWORK MANAGEMENT SYSTEM

Introduction

The Autonomous Electric Vehicle Charging Network Management System is designed to efficiently manage and optimize the operation of electric vehicle (EV) charging stations through intelligent and automated processes. With the rapid growth of electric vehicles, there is an increasing demand for a smart infrastructure that can handle charging station availability, reduce waiting times, and provide seamless user experience. This system integrates modern technologies such as IoT, cloud computing, and real-time data processing to monitor charging stations, manage reservations, and ensure efficient energy usage. By enabling communication between users, charging stations, and central management systems, it aims to create a reliable, scalable, and user-friendly EV charging ecosystem.

Objectives

The primary objective of this system is to provide an intelligent platform that manages EV charging networks efficiently and autonomously. It aims to allow users to easily locate nearby charging stations, check real-time availability, and book charging slots in advance. Another key objective is to optimize resource utilization by reducing congestion and ensuring fair allocation of charging points. The system also focuses on providing secure payment processing, real-time notifications, and accurate monitoring of charging sessions. Additionally, it seeks to support scalability to accommodate increasing users and charging stations while maintaining high performance, reliability, and security. Overall, the system strives to enhance user convenience and promote the adoption of electric vehicles through smart infrastructure management.

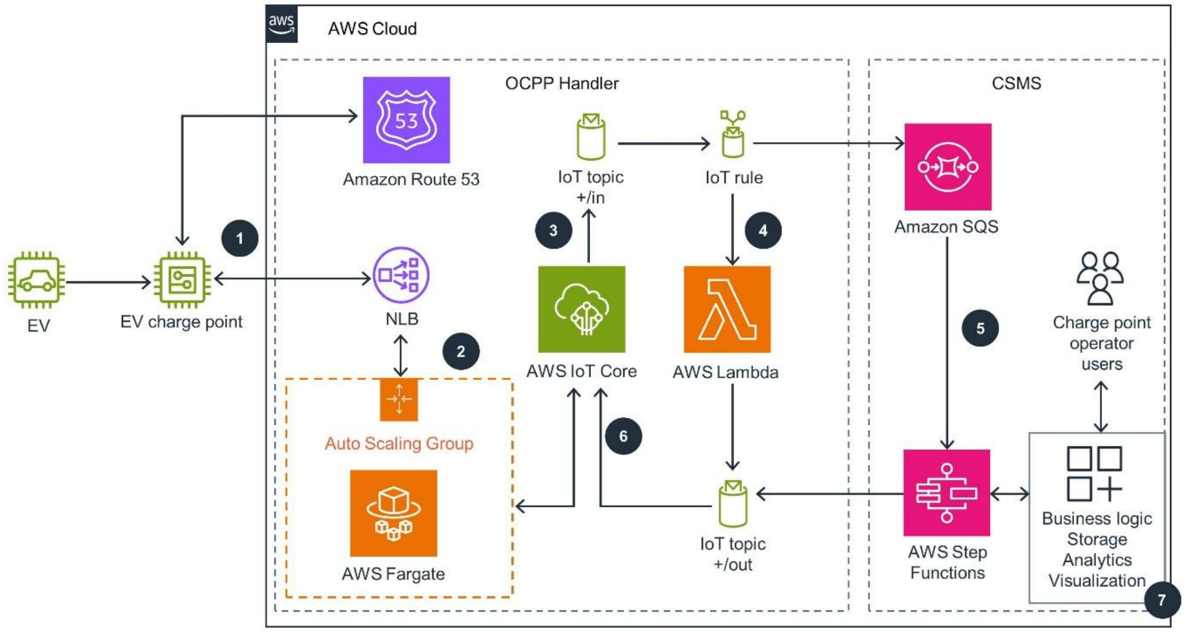
Functional Requirements

- User registration and authentication
- Search nearby charging stations (GPS-based)
- Real-time slot booking and reservation
- Charging session monitoring
- Online payment and billing
- Admin dashboard for managing stations
- Notification system (alerts, booking status)

Non-Functional Requirements

- High scalability (support thousands of users)
- Low latency response time
- Secure payment transactions
- High availability (24/7 system uptime)
- Data privacy and protection

➢ Fault tolerance and recovery



WORKFLOW:

The workflow of the Autonomous Electric Vehicle Charging Network Management System begins when a user opens the mobile or web application and searches for nearby charging stations using GPS. The system retrieves real-time data from connected charging stations and displays available slots. The user selects a preferred station and books a charging slot through the booking service. Once confirmed, the system sends a notification to the user. At the scheduled time, the vehicle connects to the charging station, and the IoT-enabled device updates the system with real-time charging status. The charging session is monitored continuously, and usage data is recorded. After completion, the payment service calculates the cost based on energy consumption and processes the transaction securely. Finally, the system generates a receipt and sends a notification to the user, while the database updates all relevant records for future reference and analysis.

Components and Their Interactions

1. User Interface (UI)

➢ Mobile/Web app for users

➢ Displays stations, bookings, payments

2. API Gateway

➢ Entry point for all client requests

- Routes requests to appropriate services

3. User Service

- Handles login, registration, authentication

4. Booking Service

- Manages slot reservation and scheduling

5. Payment Service

- Handles billing, transactions, invoices

6. Notification Service

- Sends alerts (SMS, email, app notifications)

7. Charging Station (IoT Devices)

- Provides real-time charging data
- Communicates with backend

8. Database

- Stores user data, bookings, transactions

9. Event Bus (Kafka/RabbitMQ)

- Enables real-time communication between services

Key Quality Attributes

- **Scalability** – Handle increasing EV users
- **Reliability** – Accurate booking and charging data
- **Performance** – Fast response and real-time updates
- **Security** – Secure authentication & payment
- **Maintainability** – Easy updates and debugging
- **Availability** – Continuous system access

Future Enhancements

- AI-based charging prediction system
- Integration with smart grids
- Vehicle-to-grid (V2G) support
- Renewable energy optimization

Conclusion

The Hybrid Microservices + Event-Driven Architecture is the best choice for an Autonomous EV Charging Network System. It ensures scalability, flexibility, and real-time performance while supporting future expansion and advance

d smart features. This architecture supports future advancements such as AI-based charging optimization, smart grid integration, and renewable energy management. It helps improve charging efficiency, reduces user waiting time, and encourages the adoption of electric vehicles by providing a seamless and sustainable charging experience.